M. MAATAR DATA SCIENTIST

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Profile

Ph.D. in Computer Science and Data Scientist , with 5 years of experience in R&D and software engineering. Expert in Deep Learning, Computer Vision, and NLP, coupling complex algorithm design with industrialization in scalable environments. Accustomed to international contexts (Japan, France) and demanding sectors (Offshore Construction, MedTech, Aeronautics). This profile applies expertise to design high-value bespoke solutions, ensuring the entire project lifecycle from Proof of Concept to production deployment

Skills

- **Technologies and Tools:** Python (Expert), IsaacLab, SKRL, PyTorch, Tensorboard, Numpy, Git, Docker, Pytorch-Lightning, HuggingFace, W&B, Pandas, Scikit-learn, Flask, AWS (certification in progress)
- **Domain Expertise:** Biomedical Data Analysis, Industry 4.0, Aeronautics GenAI & LLM (Fine-tuning), Computer Vision (Pose estimation), Reinforcement Learning (Robotics).
- **Methodologies and Soft-skills:** Agile Scrum, Feasibility Study, Cognitive Flexibility, Clear and Effective Communication, Collaborative Spirit and Teamwork, R&D Project Management, Advanced Algorithmics, Scientific Communication, Technical Mentoring.

Education

-PH.D. IN COMPUTER SCIENCE AND INTELLIGENT SYSTEMS | OCT. 2025 | UNIVERSITY OF AIZU

- *Topic: Neuromorphic Systems & Thermal Optimization.*

-MASTER OF SCIENCE IN DATA SCIENCE AND BIG DATA | JUNE 2021 | CY-TECH PARIS
-COMPUTER ENGINEERING DEGREE SPECIALIZED IN ERP – BI | JUNE 2018 | ESPRIT

Experience

AI/RL ENGINEER - EYES JAPAN (CLIENT: AIZAWA GROUP) | DEC. 2024 – PRESENT

Project Context: R&D and engineering project aiming to design autonomous drones capable of performing construction tasks for offshore wind farms.

Performed tasks:

- Implementation of parameterizable digital environments under NVIDIA Isaac Sim / Isaac Lab, including the definition of states and actions spaces, and agent-environment interaction mechanisms.
- Development, configuration, and training of RL policies based on PPO, SAC, and TD3, via RSL-RL and SKRL frameworks, with hyperparameters and exploration strategies optimization.
- Design and integration of cost functions, numerical penalties, and parameterizable constraints to control stability, convergence, and robustness of learning policies.
- Development of algorithmic mechanisms to sequence and progressively complexify training tasks to optimize model convergence (Curriculum Learning).

- Introduction of numerical perturbations and stochastic parameters into the calculation environment to improve the generalization capacity of trained models.
- Automation of agent, environment, cost function, and hyperparameter configuration to ensure traceability and repeatability of experiments.
- Adaptation and integration of models from simulation to a target system, with validation of policy behavior in real execution conditions.

Results: Success of the Sim2Real demonstration with validation of autonomous flight tasks in a real environment

Environments:

*Technical***:** Python, Linux, NVIDIA Isaac Sim/Lab, PPO, SAC, TD3, RSL-RL, SKRL, PyTorch, Tensorboard, Git.

*Functional***:** Offshore construction, renewable energies, offshore wind farms, automation of construction and maintenance tasks, robotics applied to complex industrial environments.

ML ENGINEER ET DATA SCIENTIST - EYES JAPAN (CLIENT: PD HOUSE ALAN) | OCT. 2022 – DEC. 2024

Project Context: Development of a MedTech Artificial Intelligence solution to objectively assess the severity of Parkinson's disease through video analysis of standardized physical exercises following the UPDRS standards.

Performed tasks:

- Development of a software pipeline integrating MediaPipe and MMPose for automated extraction of numerical representations (landmarks, time series) from videos.
- Implementation of calculation rules for the generation, normalization, and selection of features from pose data.
- Design, training, and evaluation of supervised models for classification and regression.
- Development of calculation functions to combine unit predictions into coherent and usable global scores.
- Development and deployment of interactive dashboards using Dash for feature inspection, dataset validation, and model performance monitoring

Results: Achieved 93% accuracy in classifying disease severity levels.

Environments:

Technical: Python, Linux, bash, MediaPipe, MMPose, Scikit-learn, Pandas, Numpy, Dash, OpenCV, git, plotly, Flask.

Functional: Digital Health (MedTech), neurology, medical diagnostic aid, standardized clinical assessment (UPDRS).

DATA SCIENTIST NLP & MLOPS - AIR FRANCE / KLM | DEC. 2020 – JUNE. 2021

Project Context: Industrialization project for the Data & Business Analytics Department aiming to automate the classification and processing of safety reports (Cabin Reports) to improve team responsiveness.

Performed tasks:

- Development of rules for cleaning, normalizing, and vectorizing voluminous textual corpuses.
- Fine-tuning of Large Language Models (LLMS) via selection, configuration, and training of the CamemBERT model using HuggingFace Transformers and PyTorch.

- Algorithmic adaptation of the model to the business domain by adjusting parameters and datasets to improve performance on specific terminology.
- Implementation of unit tests, writing of technical documentation, and integration of the model into internal application pipelines.

Results: ***40%*** acceleration in report processing time and reliable automation for confidence scores > ***85%***.
Environments:
Technical: Python, JupyterLab, HuggingFace (Transformers), Pandas, NumPy, PyTorch, Pytorch-Lightning (Lightning AI), Weights and Biases (W&B), Git.
Functional: Air transport, flight safety, operational risk management, semantic analysis of incident reports.

FULLSTACK DATA DEVELOPER - AIR FRANCE / KLM | DEC. 2020 - FEB. 2021

Project Context: Design and deployment of an internal web application to automate the ingestion and processing of data related to workplace accidents.
Performed tasks:

- Design and implementation of a web application using Flask, including backend API, server logic, and user interfaces.
- Development of ETL pipelines via implementation of ingestion, transformation, and validation scripts using automated calculation rules.
- Integration of the CamemBERT model for text report classification into the web service.
- Deployment of the application and support for internal teams in its usage.

Results: Deployment and adoption of the tool by business teams, eliminating repetitive processing and exchange of files and data.
Environments:
Technical: Python, Flask, Pandas, HuggingFace, Pytorch-Lightning (Lightning AI), PyTorch, Numpy.
Functional: Occupational Health and Safety (OHS), occupational risk prevention, digitalization of internal processes